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ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Course: B. Tech (Aerospace)

Name of the Paper: Aero Thermodynamics

Semester: **III**

Paper Code: **TAS 303**

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Describe the following thermodynamic systems: closed system, open system, and isolated system.	CO1
(b)	State the first law of thermodynamics. Also, give the expressions of energy balance for the following non-flow processes: i) cyclic process and ii) process undergoing a change of state.	
(c)	Describe the concept of continuum.	
Q2	(20 marks)	
(a)	Describe the steady flow energy equation, and apply it to evaluate the output velocity of a nozzle.	CO2 & CO3
(b)	A heat engine working on Carnot cycle converts one-fifth of the heat input into work. When the temperature of the sink is reduced by 80°C, the efficiency gets doubled. Make calculations for the temperature of source and sink.	
(c)	A reversed heat engine absorbs 250 kJ of heat from a low temperature region and has a mechanical input of 100 kJ. What would be the heat transferred to the high temperature region? Also evaluate the coefficient of performance of the reversed heat engine when working as a refrigerator and as a heat pump.	
Q3	(20 marks)	
(a)	Define the entropy. Also state the entropy principal.	CO4
(b)	Derive the expression for the entropy change in an irreversible process.	
(c)	Describe the available energy and the unavailable energy.	
Q4	(20 marks)	
(a)	Explain the working of an IC engine with the help of an indicator p-v diagram.	CO5
(b)	Draw the p-v and T-s diagrams for the Brayton cycle. Also write the expression for its efficiency.	
(c)	Give a comparison of Otto, Diesel and Dual cycles.	
Q5	(20 marks)	
(a)	Explain the working of a vapour compression refrigerant plant.	CO6
(b)	Describe the working of a steam power plant operating on the Rankine cycle with the help of a T-s diagram.	
(c)	Describe the saturated and superheated steam and write the expression for the dryness fraction.	